

YARKOVO, Russian Federation

WMO: 283660

Lat: 57.417N

Lon: 67.083E

Elev: 52

StdP: 100.70

Time Zone: 5.00 (E05)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.5	(c) -30.2	(d) -35.8	(e) 0.1	(f) -32.5	(g) -33.1	(h) 0.2	(i) -30.0	(j) 6.4	(k) -11.4	(l) 5.7	(m) -12.0	(n) 1.1	(o) 340	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	29.4	19.8	27.7	19.2	26.1	18.6	21.2	27.4	20.2	26.0	19.4	24.6	2.3	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.9	24.0	18.1	13.1	23.1	17.3	12.5	22.3	61.7	27.6	58.4	26.1	55.6	24.8	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 6.4	(b) 5.4	(c) 4.9	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) 6.4	(b) 5.4	(c) 4.9	(d) WB	(e) -36.6	(f) 22.5	(g) 2.5	(h) 1.4	(i) -38.4	(j) 23.5	(k) -39.8	(l) 24.3	(m) -41.2	(n) 25.1	(o) -43.1	(p) 26.2	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-16.4	-13.0	-5.6	4.6	11.1	17.1	18.3	16.4	9.9	1.8	-6.9	-13.2
	DBStd	13.47	7.40	7.40	5.79	4.81	5.35	4.42	3.61	4.04	4.07	5.23	7.19	7.79
	HDD10.0	3707	818	643	482	174	53	3	1	3	51	257	506	718
	HDD18.3	6044	1076	876	741	412	232	76	46	87	255	511	756	976
	CDD10.0	822	0	0	0	12	86	216	257	202	46	3	0	0
	CDD18.3	118	0	0	0	0	7	38	45	27	1	0	0	0
	CDH23.3	1112	0	0	0	1	102	368	416	222	4	0	0	0
	CDH26.7	256	0	0	0	0	18	98	93	47	0	0	0	0
(14) Wind	WSAvg	2.3	2.2	2.2	2.7	2.7	2.7	2.2	2.0	2.0	2.0	2.4	2.4	2.1
(15) Precipitation	PrecAvg	557	35	26	35	28	45	61	75	62	48	47	52	42
	PrecMax	684	72	56	70	57	94	127	118	107	84	112	106	88
	PrecMin	424	3	3	3	3	12	19	24	21	11	4	10	14
	PrecStd	70	18	13	18	14	21	23	28	21	19	25	21	18
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.6	3.6	7.2	21.1	29.1	31.2	31.6	30.2	23.7	18.5	6.0	1.5
		MCWB	-1.5	0.9	2.8	12.9	15.4	20.8	20.8	21.0	16.0	10.6	4.5	0.0
	2%	DB	-1.8	1.7	4.7	18.5	26.2	29.3	29.0	27.9	21.4	13.5	4.2	-0.2
		MCWB	-2.7	0.0	1.4	11.3	16.1	19.9	20.0	20.5	14.8	8.5	3.1	-0.8
	5%	DB	-3.6	-0.2	3.5	15.6	23.8	27.5	27.3	25.9	18.6	10.6	2.6	-1.7
		MCWB	-4.3	-1.6	0.8	9.9	14.3	19.3	19.3	19.5	13.2	7.7	1.7	-2.4
	10%	DB	-5.7	-2.4	2.2	12.9	21.1	25.3	25.8	23.8	16.5	8.9	1.3	-3.4
		MCWB	-6.4	-3.4	0.1	8.2	13.1	18.1	18.7	18.3	12.3	6.9	0.4	-4.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.3	1.3	3.3	14.2	18.4	23.0	22.1	22.2	17.2	11.4	5.0	0.4
		MCDB	-0.9	3.1	5.7	19.9	25.1	30.5	28.4	28.1	22.0	16.6	5.7	1.0
	2%	WB	-2.5	0.1	2.1	12.3	16.6	21.0	21.1	20.8	15.4	9.6	3.2	-0.8
		MCDB	-1.8	1.8	4.3	16.9	23.8	27.4	27.6	26.5	20.3	12.0	3.9	-0.2
	5%	WB	-4.3	-1.6	1.2	10.4	15.1	19.8	20.1	19.9	13.9	8.2	1.8	-2.4
		MCDB	-3.5	-0.4	2.9	15.1	22.4	26.0	25.9	25.0	17.5	10.4	2.5	-1.8
	10%	WB	-6.3	-3.4	0.3	8.2	13.6	18.7	19.4	19.0	12.6	6.7	0.5	-4.1
		MCDB	-5.7	-2.4	2.1	12.7	20.1	24.2	24.4	23.3	15.8	8.9	1.2	-3.4
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	8.1	9.2	9.4	11.4	10.7	10.2	10.4	8.5	6.5	6.1	6.5
		MCDBR	7.0	7.4	8.9	13.2	14.7	13.7	13.2	12.9	11.6	9.8	5.9	5.8
		MCWBR	6.8	6.3	6.4	7.7	6.7	5.9	5.5	6.4	6.1	6.3	5.3	5.5
	5% WB	MCDBR	6.8	7.1	7.8	12.0	13.5	12.1	11.4	11.8	10.7	9.4	5.3	5.9
		MCWBR	6.7	6.1	6.0	7.4	7.0	5.6	5.3	6.0	5.8	6.5	4.9	5.7
(40) Clear-Sky Solar Irradiance	taub		0.279	0.311	0.368	0.412	0.407	0.408	0.419	0.427	0.379	0.349	0.306	0.278
	taud		2.026	2.028	1.943	2.072	2.242	2.307	2.280	2.221	2.318	2.254	2.106	1.983
	Ebn at Noon		544	698	750	790	828	832	812	771	756	662	508	416
	Edh at Noon		59	93	134	139	125	119	120	119	93	75	55	47
(44) All-Sky Solar Radiation	RadAvg		0.69	1.67	3.16	4.43	5.23	5.70	5.42	4.19	2.63	1.32	0.62	0.40
	RadStd		0.05	0.12	0.20	0.36	0.36	0.40	0.50	0.40	0.28	0.20	0.10	0.06

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page